

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1898

South Meadow 138 kV
September 26, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1898. The Interconnection Customer proposes a 105.2 MW solar photovoltaic facility interconnecting at the South Meadow 138 kV substation in the Pennsylvania area. The request was received on October 06, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 671 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,063,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
TARPON SPRINGS / LA CANADA 138.0 kV	CEDAR CRK / YORKANA 345.0 kV	181.9	200	6.0
LEVEE / ZAHARY 138.0 kV	TALENERGY MARTINS CREEK LLC HARRISBURG 500 kV/345 kV XFMR	153.1	215	6.8
HAMILTON PA / KNOXVILLE BATTERY ENERGY STORAGE 138.0 kV line	METALCAN / KANSAS RIVER PROJECT 138.0 kV	143.7	166	9.9
S MEADOW SUBSTATION / MONROE NE 345.0 kV	CHANNEL ENERGY CENTER / MONROE NE 500.0 kV	140.8	1,187	37.2
OTTEN / KNOXVILLE BATTERY ENERGY STORAGE 138.0 kV Line	METALCAN / KANSAS RIVER PROJECT 138.0 kV	128.3	194	9.9
YORKANA / SAFE HARBOR 345.0 kV	BRUNNER ISLAND / BIXBY HILL 500.0 kV	118.1	1,216	13.2
ZACHARY / HAMILTON PA 138.0 kV	KANSAS RIV PROJECT / METALCAN 138.0 kV	114.5	208	9.9
NORWAY IN / LEVEE 138.0 kV	J STREET / LEVEE 138.0 kV	106.5	200	5.9
POWER DELIVERY PT / TOLEDO 138.0 kV	MCCLELLAN-ADC SUBSTATION / DEERFIELD 5 138.0 kV	106.5	234	5.9
CEDAR CRK / YORKANA 345.0 kV	CO LINE RD / MONROE NE 500.0 kV	104.9	1,238	25.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TARPON SPRINGS / LA CANADA 138.0 kV	Rebuild line with higher-capacity conductor	\$2,227,000	8% (\$168,000)
LEVEE / ZAHARY 138.0 kV	Rebuild line with higher-capacity conductor	\$16,448,000	78% (\$12,746,000)
HAMILTON PA / KNOXVILLE BATTERY ENERGY STORAGE 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,433,000	100% (\$2,433,000)
S MEADOW SUBSTATION / MONROE NE 345.0 kV	Rebuild line with higher-capacity conductor	\$27,332,000	5% (\$1,487,000)
OTTEN / KNOXVILLE BATTERY ENERGY STORAGE 138.0 kV Line	Rebuild line with higher-capacity conductor	\$2,756,000	100% (\$2,756,000)
SAFE HARBOR / YORKANA SUBSTATION 345.0 kV	Rebuild line with higher-capacity conductor	\$67,095,000	2% (\$1,172,000)
ZACHARY / HAMILTON PA 138.0 kV	Rebuild line with higher-capacity conductor	\$2,008,000	100% (\$2,008,000)
NORWAY IN / LEVEE 138.0 kV	Reconductor line	\$2,523,000	100% (\$2,523,000)
POWER DELIVERY PT / TOLEDO 138.0 kV	Reconductor line	\$3,045,000	21% (\$645,000)
CEDAR CRK / YORKANA 345.0 kV	Reconductor line	\$5,173,000	2% (\$125,000)
	Total allocated to project		\$26,063,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.